

AIDE/IPDE PHASE I-II TITE SIMULATOR REBUILD

Code architecture and function interface specification

Prepared from the current Phase I-II implementation, the prior TITE event-clock helper, the CRM helper, the four efficacy JAGS models, and the August 2026 trial-design specification.

Item	Specification
Document type	Implementation blueprint; no R source code is replaced by this document
Target engine	One-stage and two-stage AIDE/IPDE with event-clock TITE operation
TITE endpoints	Model-specific TITE toxicity and delayed-outcome efficacy; common T_assess window
Queue policy	New patients first
Escalation gate	n_eval blocks escalation only; a triggering new arrival is dropped before queue entry, while a triggering retreat opportunity remains queued
Date	August 6, 2026

Purpose and specification hierarchy

Purpose. This document defines how to rebuild the Phase I-II simulator by combining the reusable event-clock mechanics in AIDE_CRM_helper_TITE.R with the model, utility, monitoring, one-stage, and two-stage components in AIDE_phase_I_II(2).R and AIDE_CRM_helper_modified(1).R. It describes the target file structure, state objects, event loop, required functions, and the input/output contract for every major component.

Authority order. When existing code and older text conflict, the rebuild should use the following order: (1) the user's latest instructions, (2) Appendix C of IPDE_Design_with_Efficacy_Event_Loop_n_eval_Updated.pdf, (3) the earlier sections of that design document, and (4) legacy code behavior. This is necessary because the main design text still contains older directional and enrollment options that Appendix C later supersedes.

- A cohort that is already open is filled at its frozen next dose; no new model or design decision is made while slots remain open.
- When no cohort is open, every assignable new-patient or retreat event triggers an immediate model update and application of the prespecified one-stage or two-stage algorithm.
- The `n_eval` rule is an escalation-only gate. It does not delay a stay or de-escalation decision.
- If the algorithm recommends stay or de-escalation, the cohort opens immediately at the returned next dose and is filled from the queue.
- If the algorithm recommends escalation and fewer than `n_eval` administrations have completed DLT evaluation at the current dose, the escalation is blocked. If the triggering event is a new-patient arrival, that patient is dropped and is never added to the waiting queue. If the triggering event is a retreat opportunity, the retreat patient remains in the queue for consideration at a future assignable event. All other queued opportunities are retained.
- DLT and efficacy use the same assessment window `T_assess`. A retreat opportunity is created only after that window is complete, with no DLT, no efficacy response, and cycle below `cycle_max`.
- There is one cohort next dose. New and retreat patients filling that cohort receive the same model-and-design-selected dose; no separate retreat destination is selected.

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1. Source files and current responsibilities

Source	Role in rebuild	What to reuse
AIDE_CRM_helper_TITE.R	Reusable event-clock pattern	Contains <code>future_events</code> , waiting queue, <code>cohort_open/cohort_enrolled</code> state, event scheduling, TITE toxicity data reconstruction, slot-by-slot cohort filling, decision logging, and an operating-characteristic wrapper.

Source	Role in rebuild	What to reuse
AIDE_phase_I_II(2).R	Current Phase I-II scientific logic	Contains one-stage and two-stage allocation rules, four efficacy carryover models, TITE toxicity/efficacy data builders, futility, utility, MTD/OBD selection, recycle gates, trial simulation, and OC aggregation.
AIDE_CRM_helper_modified(1).R	Toxicity model adapter	Contains unified CRM data preparation, TITE weights, fit/move/final-selection functions, and multiple CRM backends. The rebuilt TITE path should dispatch to the toxicity model selected in config.
fix_CRM_TITE(1).bug	Reference TITE toxicity likelihood	Implements $\theta_{\text{eff}} = w * \theta_{\text{base}}$ for one fixed-r model. Reuse its likelihood pattern as an example; it must not define the only TITE toxicity backend.
beta_binomial_tite_dose_specific_carryover.jags	E-RD efficacy model	Dose-specific complement-discount $r_{E,j}$ with delayed nonresponse contribution through the zeros trick.
beta_binomial_tite_shared_carryover.jags	E-RS efficacy model	Shared complement-discount r_E with delayed nonresponse contribution.
previous_dose_additive_dose_specific_tite_beta_binomial_efficiency.jags	E-AD efficacy model	Destination-dose-specific additive $\alpha_{E,j}$ using the patient's previous dose.
previous_dose_additive_tite_beta_binomial_efficiency.jags	E-AS efficacy model	Shared additive α_E using the patient's previous dose.

1.1 Reuse strategy

- Reuse the old TITE helper as the event-engine template, not as the scientific decision rule. Its event table, queue, cohort-open state, and slot-filling pattern are useful, but its fixed-r restriction and its old suspension behavior must be replaced.
- Reuse the current Phase I-II model-fitting, utility, monitoring, and stage-allocation code only after extracting it from the synchronous cohort loop into testable decision functions.
- Reuse the CRM helper as a model adapter. The event engine should consume a standardized model result rather than inspecting backend-specific fields throughout the loop.
- Reuse the supplied efficacy JAGS files as model-specific TITE backends through a common R wrapper. MCMC draws may be used transiently inside a fit, but they are discarded after the required posterior summaries and decision probabilities are computed.

2. Gap analysis: current implementation versus target design

Area	Current behavior	Required behavior
Event loop	AIDE_phase_I_II enrolls a whole cohort synchronously, then immediately proceeds to the next model step.	Use an explicit future-event table, waiting queue, cohort-open flag, and slot-by-slot filling.

Area	Current behavior	Required behavior
Decision timing	Current TITE Phase I-II does not implement the closed-cohort n_eval gate.	At every closed-cohort assignable event, update the selected models and run the one-stage or two-stage algorithm immediately.
Scope of n_eval	The old TITE helper uses n_eval_escalate only to block escalation.	n_eval is checked only when the returned next dose is above the current dose. Stay and de-escalation are never delayed by n_eval.
Queue retention	Legacy helper can drop arrivals during suspension; current Phase I-II has no unified retained queue.	If n_eval blocks a proposed escalation, drop the triggering new-patient arrival without adding it to the queue. If the trigger is a retreat opportunity, retain it in the queue for future use. Retain all other waiting opportunities.
Priority	Current Phase I-II supports continuous or IPDE-first schemes.	For the rebuilt TITE strategy, waiting new patients are always considered before retreat events.
Recycle timing	Current TITE code requires both toxicity and efficacy windows to be complete.	Use one common T_assess window. Retreat eligibility requires completion of that window with no DLT and no efficacy response.
Recycle direction	Current aide_phase12_ipde_dose_eligible restricts destinations using ipde_design and flexible_ipde.	Remove separate retreat-destination rules. The design algorithm returns one next dose for the cohort.
Cohort dose	Current enroll_cohort assigns every administration in a cohort to one dose.	Every administration in a cohort, whether new or retreat, receives the same next dose.
Model ownership	Some recycle logic is named and activated only for random-r CRM.	The selected toxicity and efficacy models determine the model summaries used by the design algorithm; there are no separate destination-specific recycle gates.
Model pairing	carryover_model maps one toxicity choice to one efficacy choice.	Select toxicity and efficacy models independently: two toxicity options and four efficacy options. TITE uses the selected model for each endpoint and does not force fixed r.
Stage I threshold	Current code uses N_s1 at the current dose.	The design document defines s1_Max as cumulative Stage I administrations plus a stay recommendation.

Area	Current behavior	Required behavior
Futility update	Current code persistently assesses mainly the just-treated dose.	At each model update, return a full per-dose futility vector based only on fully ascertained efficacy outcomes. Pending efficacy outcomes are excluded from the futility numerator, denominator, and minimum-information count.
DGM versus analysis	Current true IPDE efficacy generation is additive even when another analysis model is selected.	Restrict both endpoint DGMs to $q_j = \min(1, p_j + \alpha_{\text{true}} p_{\{j-1\}})$, with $p_0 = 0$; keep DGM parameters separate from analysis-model parameters.

2.1 Key architectural consequence

The rebuilt simulator should separate the event engine from the scientific decision functions. The event engine determines when a closed-cohort assignable event occurs and which queued patient fills a slot. The selected toxicity and efficacy adapters update the models, and the one-stage or two-stage algorithm returns one next dose for the entire cohort. Retreat eligibility is structural; it does not create a second dose-selection problem.

3. Proposed code structure

A modular layout is recommended so that event logic, statistical models, allocation algorithms, and reporting can be tested independently. Existing public function names can remain as compatibility wrappers.

Target file/module	Responsibility
R/aide_phase12_config.R	Configuration constructors, validation, model-name normalization, file resolution.
R/aide_phase12_state.R	State constructors and validators for events, queue, cohort, administrations, and logs.
R/aide_phase12_endpoints.R	Binary event-time generation and interim toxicity/efficacy data reconstruction.
R/aide_phase12_toxicity.R	Selected toxicity-model adapters, overtotoxicity, model-specific TITE likelihoods, recommendations, and final MTD.
R/aide_phase12_efficacy.R	Four efficacy-model adapters, delayed-outcome likelihood with T_assess, futility, and posterior summaries without stored draws.
R/aide_phase12_utility.R	Utility 1-3 calculation and conservative tie-breaking.
R/aide_phase12_design.R	One-stage and two-stage decision functions; stage transition, candidate sets, action, and one next dose.
R/aide_phase12_recycle.R	Retreat structural eligibility and queue/assignment audit; no separate destination selection.
R/aide_phase12_events.R	Event scheduling, queue operations, cohort opening/filling, escalation-only n_eval gate, and main event loop.

Target file/module	Responsibility
R/aide_phase12_final.R	Final MTD/OBD analysis and standardized trial summary.
R/aide_phase12_oc.R	Replicate simulation, aggregation, reproducible seeding, and optional raw trial tables without posterior draws.
inst/jags/	Model-specific toxicity and efficacy TITE files. The fixed-r file is one reference backend, not a restriction.
tests/testthat/	Unit tests for event order, model adapters, algorithms, recycling, and final outputs.

3.1 Dependency direction

```
simulate_AIDE_phase_I_II_event()
-> event engine
  -> closed-cohort assignable event
  -> model-specific TITE or complete-outcome adapters
    -> selected toxicity backend
    -> selected efficacy backend
  -> one-stage or two-stage decision
    -> one action: de-escalate, stay, escalate, or stop
    -> one next dose
  -> escalation-only n_eval gate
    -> if blocked and trigger is new: drop before queue entry
    -> if blocked and trigger is retreat: retain in queue
  -> cohort opening and queue filling
    -> new patient first
    -> structurally eligible retreat second
    -> all assignments use the same next dose
  -> endpoint generation and future-event scheduling
-> final MTD/OBD analysis
-> standardized logs and summary-only model records
```

Compatibility rule. Keep `simulate_AIDE_phase_I_II()` as a thin wrapper that translates the old long argument list into a validated config and scenario object, then calls the new event engine. New internal functions should not depend on variables captured from a 1,000-line parent function.

4. Core data structures and invariants

4.1 Configuration object

Sub-list	Required fields
design	allocation, cohort_size, Nmax, s1_Max, N_s2, n_eval, m_U, cycle_max, start_dose
time	arrival_rate, t0, T_assess, and DLT/response event-time distributions
toxicity	selected analysis model (discount-r or additive-alpha), target, cutoff, skeleton, priors, and MCMC controls
efficacy	one of four models, regular prior, carryover prior, MCMC controls, futility threshold/cutoff
utility	utility_type, lambda_T, four-outcome scores, tie-break rule

Sub-list	Required fields
recycle	new-first priority, structural retreat eligibility, cycle limit, and type-specific blocked-escalation policy: drop new trigger, retain retreat trigger
reporting	store event/queue/decision tables, verbose, and optional raw trial-table storage; posterior draws and model diagnostics are never stored

4.2 Scenario object

The analysis model and the data-generating mechanism remain separate. However, both toxicity and efficacy IPDE generation are restricted to the adjacent-lower-dose additive form $q_j = \min\{1, p_j + \alpha_{\text{true}} p_{j-1}\}$, with p_0 defined as 0 for the boundary at dose 1. The generated IPDE probability depends on the assigned dose j , not on a selectable DGM type or the patient's actual previous dose.

Field	Meaning
p_true	Length-J regular toxicity probabilities.
e_true	Length-J regular efficacy probabilities.
toxicity_ipde_dgm	$\alpha_{\text{true},T}$; generate $q_{T,j} = \min(1, p_{\text{true}}[j] + \alpha_{\text{true},T} * p_{\text{true}}[j-1])$, with $p_{\text{true}}[0] = 0$.
efficacy_ipde_dgm	$\alpha_{\text{true},E}$; generate $q_{E,j} = \min(1, e_{\text{true}}[j] + \alpha_{\text{true},E} * e_{\text{true}}[j-1])$, with $e_{\text{true}}[0] = 0$.
endpoint_time_dgm	Common T_{assess} plus DLT and efficacy event-time distribution controls.

4.3 Event table

Column	Contract
event_id	Unique integer event identifier.
time	Calendar time.
seq	Monotone sequence number for deterministic tie-breaking.
event_type	new_arrival, retreat_available, dlt_observed, efficacy_response, assessment_complete, or internal_stop.
patient_id	Patient linked to the event; NA only for global internal events.
admin_id	Administration that generated the event, when applicable.
processed	Logical or status field if events are retained rather than popped.

4.4 Waiting queue

The queue contains retained assignment opportunities, not model-update events. New-patient and retreat entries use event-time/sequence ordering, with new patients selected first whenever a cohort is open. A retreat entry is assignable only after structural eligibility is met. For a closed-cohort decision attempt, the triggering new-patient arrival is held transiently and is not added to the queue until the decision permits cohort opening or another rule explicitly retains it. If `n_eval` blocks escalation, that new patient is dropped without queue entry. A triggering retreat opportunity remains in the queue for future use. All other waiting entries are retained.

Column	Contract
<code>queue_id</code>	Unique queue row.
<code>time, seq</code>	Original assignable-event order.
<code>type</code>	new or retreat.
<code>patient_id</code>	Patient identifier.
<code>source_admin_id</code>	Prior administration for retreat; NA for new.
<code>status</code>	waiting, assigned, dropped_new_n_eval_block, retained_retreat_n_eval_block, removed_ineligible, or trial_ended.
<code>last_reason</code>	Most recent reason not assigned, including not structurally eligible, new trigger dropped before queue entry, or retreat trigger retained after blocked escalation.

4.5 Administration table

Column group	Contract
<code>admin_id, patient_id</code>	Unique administration and patient IDs.
<code>cohort_id, decision_id</code>	Cohort and decision that authorized the assignment.
<code>stage</code>	stage1, stage2, or one_stage.
<code>assignment_type</code>	new or retreat.
<code>dose</code>	Single dose selected for the cohort and received by this administration.
<code>decision_next_dose</code>	Frozen next dose from the decision that opened the cohort; must equal dose.
<code>previous_dose</code>	Immediately preceding administered dose for retreat; sentinel/NA for a new patient.
<code>cycle</code>	Administration number for the patient.
<code>t_arrival, t_start</code>	Patient arrival and administration start time.
<code>t_dlt, t_response</code>	Event times when the generated DLT or efficacy response occurs; Inf/NA when absent.
<code>assessment_end</code>	Common endpoint-window end <code>t_start + T_assess</code> .

Column group	Contract
dlt_final, eff_final	Generated complete outcomes.
true_p_tox, true_p_eff	Row-specific DGM probabilities for traceability.

4.6 Cohort state

Field	Contract
open	TRUE while fewer than cohort_size slots have been assigned.
cohort_id	Current cohort identifier.
opened_time	Time of the model/design decision.
next_dose	Single dose assigned to every new and retreat patient in the cohort.
decision_action	de-escalate, stay, or escalate action that produced next_dose.
stage	Active stage at cohort opening.
capacity, filled	Total slots and number assigned.
decision_id	Decision record that opened the cohort.

4.7 State invariants

- No model-based dose decision occurs while cohort\$open is TRUE.
- Every patient assigned to an open cohort, whether new or retreat, receives cohort\$next_dose.
- A retreat patient requires only structural eligibility: completed T_assess, no DLT, no efficacy response, and cycle < cycle_max. No separate destination-specific toxicity or efficacy gate is applied.
- The initial cohort is opened by initialization at start_dose and is exempt from the n_eval escalation gate.
- When cohort\$open is FALSE, each assignable event runs the selected models and design algorithm. The n_eval check is applied only if the returned action is escalation.
- Every administration counts toward Nmax, whether the assignment is new or retreat.
- A patient has at most one waiting retreat opportunity for the same completed administration.

5. Event-clock control flow

```

initialize state, future events, queue, and starting cohort
fill the starting cohort when assignment opportunities are available

while trial is active and administration cap not reached:
  event <- pop earliest event by time then sequence
  advance t_now and update endpoint status implied by event

  if event completes T_assess with no DLT and no efficacy response
  and the patient remains below cycle_max:
    create or confirm one retreat opportunity in the queue

  if a cohort is open:
    if event is a new arrival:
      add the new-patient opportunity to the queue
      fill as many open slots as possible
      - select the earliest waiting new patient first
      - then select the earliest structurally eligible retreat
      - assign every selected opportunity to cohort$next_dose
    continue to the next event

  # no cohort is open: this assignable event triggers a decision
  # a new arrival is transient and not yet queued; a retreat is retained in queue
  interim <- rebuild toxicity and efficacy data at t_now
  fits <- update the selected toxicity and efficacy models
  decision <- apply monitoring and the active one-stage/two-stage algorithm

  if decision stops the trial:
    terminate
  if decision has no assignable next dose:
    leave the cohort closed and retain existing queue entries
    continue

  if decision$action == "escalate":
    n_current <- count evaluated administrations at the current dose
    if n_current < n_eval:
      if event$type == "new":
        drop the triggering new patient without adding to queue
        log trigger_disposition = "new_dropped_n_eval"
      else if event$type == "retreat":
        keep the triggering retreat opportunity in the queue
        log trigger_disposition = "retreat_retained_n_eval"
      leave the cohort closed
      continue

  if event$type == "new":
    add the triggering new-patient opportunity to the queue
    open a cohort at decision$next_dose
    fill available slots immediately from the queue, new first

complete the endpoint follow-up required for final analysis
select final MTD and OBD
return data tables, logs, and posterior summaries without stored draws or model diagnostics

```

5.1 Escalation-only n_eval gate

n_eval counts administrations at the current dose whose DLT is observed by t_now or whose common assessment window T_assess has ended without DLT. Each qualifying administration counts once, including repeated administrations from the same patient. The count is consulted only after the model-and-design decision proposes a higher dose. If the count is below n_eval , the escalation is blocked. A triggering new-patient arrival is dropped and never added to the queue. A triggering retreat opportunity remains in the queue and may be used at a future assignable event.

5.2 Open-cohort behavior

- An endpoint update while a cohort is partially filled may change recorded outcome status, but it does not change the next dose frozen when that cohort opened.
- Later assignable events fill the remaining slots at the same next dose without another model decision.
- The cohort closes exactly when filled reaches $cohort_size$. A cohort is dose-homogeneous even when it contains both new and retreat patients.

6. Required functions and input/output contracts

The tables below list the required testable functions. Names are recommendations; compatibility wrappers may preserve existing public names. “Reuse” means the existing function can be retained with minor validation changes. “Modify” means its scientific core is useful but its interface or logic must change. “New” means a separate function should be created rather than embedded as a nested closure.

6.1 Configuration and state

Function	Status	Inputs	Output	Responsibility
aide_phase12_config())	New	Design/model/time/ monitoring arguments.	Validated config list with normalized model names and resolved files.	Construct one authoritative configuration.
aide_phase12_scenar io()	New	p_true, e_true, alpha_true for toxicity and efficacy IPDE DGMs, and endpoint- time parameters.	Validated scenario list using only adjacent- lower-dose additive IPDE generation.	Separate true DGM from analysis model.
aide_phase12_validat e_config()	New/derive	config, scenario.	Invisible TRUE or normalized objects; informative error.	Cross-field checks, including one common T_assess and availability of the selected model- specific TITE backend.
aide_phase12_initializ e_state()	New	config, scenario, seed.	State list with events, queue, cohort, admin, logs, stage, elimination vectors.	Create reproducible initial trial state and starting cohort.
aide_phase12_validat e_state()	New	state, config.	Invisible TRUE or error.	Check invariants during debug/test runs.

6.2 Event and queue primitives

Function	Status	Inputs	Output	Responsibility
aide_event_schedule())	Modify from old helper	events, time, event_type, patient_id, admin_id.	Updated events table.	Append event with monotone sequence.
aide_event_pop_next())	Modify	events.	list(event, remaining_events).	Pop earliest time/sequence event.
aide_queue_add()	New	queue, assignable event.	Updated queue.	Add new/retreat entry exactly once.
aide_queue_next_ne w()	New	queue.	Row index or integer(0).	Return earliest waiting new patient.

Function	Status	Inputs	Output	Responsibility
aide_queue_next_retre eat()	New	queue, state, config.	Row index or integer(0).	Return earliest structurally eligible retreat after new- patient entries are exhausted.
aide_fill_open_cohort()	Modify from old helper	state, t_now, config, scenario.	Updated state.	Fill slots new-first at cohort\$next_dose; no new dose decision while open.
aide_open_cohort()	New	state, design decision, t_now.	Updated state with open cohort.	Freeze next dose, action, stage, and decision ID.

6.3 Endpoint generation and interim data

Function	Status	Inputs	Output	Responsibility
aide_draw_binary_ev ent_time()	Generalize	probability, start_time, window, dist, alpha.	outcome, event_time, window_end.	One common endpoint-time generator for DLT and response.
aide_append_adminis tration()	New	state, patient, dose, type, stage, DGM probabilities.	Updated state and admin_id.	Add administration and schedule endpoint events.
aide_phase12_tite_to xicity_data()	Reuse	admin, t_now, T_assess.	Interim toxicity rows with observed y, w_T, prior dose, and IPDE indicator.	Prepare input for the selected model- specific TITE toxicity backend.
aide_phase12_tite_eff icacy_data()	Modify	admin, t_now, T_assess, ndose.	Rows with observed response, ascertainment indicator, w_E, previous dose.	Prepare delayed- outcome efficacy input using the same T_assess as toxicity.
aide_build_interim_da ta()	New	state, t_now, config.	list(toxicity, efficacy).	Single source of truth for each model update.
aide_count_evaluated _current_dose()	New	admin, current_dose, t_now, T_assess.	list(n, qualifying_admin_ids) .	Count evaluated administrations for the escalation-only n_eval gate; repeated administrations from the same patient count separately.

6.4 Toxicity model interface

Function	Status	Inputs	Output	Responsibility
crm_prepare_tite_weight()	Reuse	data and endpoint timing/weight fields.	Weight vector in [0,1].	Common TITE weight handling.
crm_fit()/crm_fit_discount()	Reuse through adapter	toxicity data, selected r/alpha model, skeleton, priors, MCMC, TITE controls.	Transient backend fit and required posterior summaries; sample matrices discarded before return.	Fit the selected toxicity model in complete-outcome or TITE mode.
aide_fit_toxicity()	New adapter	interim toxicity data, config.	Standard toxicity_fit summary object without posterior draws.	Hide BOIN/CRM/backend-specific field names.
aide_toxicity_recommendation()	Modify crm_move logic	current dose, toxicity_fit, counts, config, eliminated.	action, recommended dose, stop flag.	Return toxicity recommendation only; do not open cohort.
aide_apply_escalation_gate()	New	decision action/next dose, admin, current dose, t_now, n_eval, trigger queue row.	allowed/blocked, evaluated administration count/IDs, trigger type/disposition, new-patient drop flag, retreat-retained flag, and updated queue status.	Apply n_eval only to escalation; drop a triggering new arrival before queue entry and retain a triggering retreat opportunity in the queue.
aide_final_toxicity_analysis()	Adapt select.mtd.crm	complete final toxicity data, config.	MTD, p_T, elimination, posterior summary.	Final toxicity inference.

6.5 Efficacy model interface

Function	Status	Inputs	Output	Responsibility
aide_phase12_expanded_beta_prior()	Reuse	prior, ndose, name.	ndose x 2 matrix.	Normalize shared/dose-specific prior inputs.
aide_phase12_tite_efficiency_posterior()	Reuse/standardize	interim efficacy rows, model option, priors, MCMC.	Per-dose posterior summaries needed by the design; MCMC sample matrix discarded.	Fit one of four delayed-outcome efficacy models using T_assess.
aide_fit_efficacy()	New adapter	interim efficacy data, config, cache.	Standard efficacy_fit summary object without posterior draws or model diagnostics.	Expose p_E means, carryover means, IPDE means, and decision probabilities only.

Function	Status	Inputs	Output	Responsibility
<code>aide_update_futility()</code>	Modify	fully ascertained efficacy rows, previous persistent vector, efficacy threshold/cutoff, <code>min_eff_n_for_futility</code> .	Updated length-J futility vector, fully ascertained administration counts, and posterior futility probabilities.	Evaluate every eligible dose using fully ascertained efficacy outcomes only; exclude pending efficacy rows entirely.
<code>aide_efficacy_ipde_summary()</code>	New adapter output	transient efficacy fit, model option, <code>ndose</code> .	Length-J posterior mean IPDE efficacy vector and carryover summaries.	Provide model-specific summaries for reporting; not a separate recycle gate.
<code>aide_compute_utility()</code>	Reuse	<code>p_E</code> , <code>p_T</code> , utility settings.	Length-J utility vector.	Utility 1-3.

6.6 Design algorithms

Function	Status	Inputs	Output	Responsibility
<code>aide_build_admissible_set()</code>	New	<code>toxicity_fit</code> , <code>efficacy_fit</code> , persistent eliminations, tried-dose/stage restrictions.	Integer candidate dose vector plus exclusion reasons.	Centralized candidate-set definition used internally by the design algorithm.
<code>aide_one_stage_decision()</code>	Extract/modify	current dose, toxicity recommendation, admissible set, <code>p_T</code> , <code>p_E</code> , utility, <code>m_U</code> quantity.	Standard <code>design_decision</code> with action and one next dose.	Apply de-escalation-first and local utility candidate sets.
<code>aide_two_stage_decision()</code>	Extract/modify	stage state, cumulative Stage I <code>n</code> , toxicity recommendation, admissible set, <code>p_E</code> , config.	Standard <code>design_decision</code> with action, one next dose, and stage transition.	Stage I toxicity allocation and Stage II efficacy maximization.
<code>aide_make_design_decision()</code>	New dispatcher	state, <code>toxicity_fit</code> , <code>efficacy_fit</code> , config.	<code>design_decision</code> .	Run monitoring, stopping, and the active algorithm; return action and one next dose.

6.7 Recycling

Function	Status	Inputs	Output	Responsibility
<code>aide_retreat_structural_eligibility()</code>	New/replace <code>eligible_ipde_ids</code>	patient latest state, <code>t_now</code> , <code>T_assess</code> , <code>cycle_max</code> .	eligible flag and reason codes.	Completed common <code>T_assess</code> ; no DLT; no efficacy response; below cycle limit.

Function	Status	Inputs	Output	Responsibility
aide_queue_retreat_opportunity()	New	completed administration, patient state, t_now, config.	Updated queue plus eligibility reason.	Create one retreat opportunity exactly once after structural eligibility is met.
aide_select_assignment_opportunity()	New	queue, state, cohort, config.	Selected queue row or none.	Select waiting new patient first, then earliest structurally eligible retreat.
aide_assign_opportunity_to_cohort()	New	selected queue row, cohort\$next_dose, state, scenario.	Updated state and administration ID.	Assign new or retreat patient to the same frozen cohort dose and schedule outcomes.
aide_log_retreat_assignment()	New	retreat opportunity, eligibility, queue status, cohort, assigned dose.	Updated retreat audit log.	Record waiting, retention after blocked escalation, assignment, and terminal status without destination tables.

6.8 Trial, final analysis, and OC wrapper

Function	Status	Inputs	Output	Responsibility
simulate_AIDE_phase_I_II_event()	New core	config, scenario, seed.	aide_phase12_trial result.	Run one event-clock trial.
simulate_AIDE_phase_I_II()	Compatibility wrapper	Existing arguments.	Same standardized result; optional legacy aliases.	Translate legacy arguments and call new core.
aide_final_analysis()	New coordinator	completed state, config.	MTD, OBD, posterior summaries required for the final decision, and no-selection reason; no sample matrices or model diagnostics.	Final sequential MTD/futility/utility selection.
aide_summarize_trial()	New	state, final analysis.	Stable public result structure.	Assemble logs and reporting metrics.
get_oc_sim_AIDE_phase_I_II()	Modify	scenario, config, ntrial, seed, store_raw_tables.	OC list/data frame and optional raw event/admin/log tables without posterior draws or model diagnostics.	Replicate trials with deterministic trial-level seeds.

6.9 Standard model output contracts

```

toxicity_fit <- list(
  model, p_regular_mean, carryover_mean, p_ipde_mean,
  prob_overtox_by_dose, eliminated
)

efficacy_fit <- list(
  model, p_regular_mean, carryover_mean, p_ipde_mean,
  prob_below_threshold, futile,
  n_fully_ascertained_by_dose
)

design_decision <- list(
  stop_trial, stop_reason, stage, stage_transition,
  current_dose,
  action,                      # de-escalate, stay, escalate, or stop
  next_dose,                   # NA means no cohort opens
  candidate_doses,             # algorithm audit only; not retreat destinations
  exclusion_reasons,
  p_toxicity, p_efficacy, utility,
  n_eval_required, n_eval_observed, escalation_blocked,
  trigger_type, trigger_disposition,
  new_patient_dropped, retreat_opportunity_retained
)

```

7. Model and algorithm integration

7.1 Toxicity

- TITE changes the likelihood contribution, not the selected toxicity model. For each administration, the active toxicity model first defines its row-specific probability $\theta_{T,i}$. Under a discount-r model, $\theta_{T,i}$ follows that model's regular/IPDE form; under an additive-alpha model, $\theta_{T,i}$ uses the current and required carryover terms. The TITE contribution then uses $w_{T,i}$ $\theta_{T,i}$ for the Bernoulli event probability, so a pending non-DLT contributes $1 - w_{T,i}$ $\theta_{T,i}$.
- The event engine does not interpret fixed r , random r , or additive α . It asks the toxicity adapter for the summaries, elimination state, and toxicity recommendation required by the design algorithm.
- Complete-outcome and TITE modes use the same model-selection configuration. The TITE flag dispatches to the time-weighted likelihood for the selected toxicity model instead of forcing the fixed-r backend.

7.2 Efficacy

Design label	R model name	IPDE probability	JAGS file
E-RD	dose_specific_carryover	$r_{E,j} + (1-r_{E,j})p_{E,j}$	beta_binomial_tite_dose_specific_carryover.jags
E-AD	dose_specific_previous_dose_additive	$\min(1, p_{E,j} + \alpha_{E,j} p_{E,l})$	previous_dose_additive_dose_specific_tite_beta_binomial_efficacy.jags
E-RS	shared_carryover	$r_E + (1-r_E)p_{E,j}$	beta_binomial_tite_shared_carryover.jags
E-AS	previous_dose_additive	$\min(1, p_{E,j} + \alpha_E p_{E,l})$	previous_dose_additive_tite_beta_binomial_efficacy.jags

Delayed nonresponse rule. An ascertained response contributes $\log(\theta_{E,E})$; an ascertained nonresponse contributes $\log(1-\theta_{E,E})$; and a pending nonresponse contributes $w_{E,E} \log(1-\theta_{E,E})$, where $w_{E,E}$ uses the same T_{assess} window as toxicity. All four efficacy models remain selectable in TITE mode. Posterior samples are used only transiently to calculate means and decision probabilities and are not stored in the trial state or returned result. Futility monitoring is separate from this delayed-outcome likelihood: only fully ascertained efficacy outcomes are included in the futility

numerator, denominator, and minimum-information count. An administration with pending efficacy status is excluded completely from the futility check.

7.3 One-stage decision

1. At each closed-cohort assignable event, update the selected toxicity and efficacy models and obtain the toxicity recommendation.
2. If direct de-escalation is required, return the de-escalated next dose immediately; utility cannot override it and `n_eval` is not checked.
3. For stay with the active current-dose sample quantity at least `m_U`, rank admissible doses in $\{d-1, d\}$; a returned stay dose is processed immediately.
4. For escalation, or stay with the active quantity below `m_U`, rank admissible doses in $\{d-1, d, d+1\}$. After a single next dose is selected, classify any `next_dose > current_dose` as escalation and apply `n_eval`.
5. Return one next dose for the cohort. Both new and retreat patients filling the cohort receive this dose.

7.4 Two-stage decision

1. Stage I allocation is toxicity-driven.
2. Stage I ends only when cumulative Stage I administrations reach `s1_Max` and the next toxicity recommendation is stay.
3. At every permitted decision time, update overtoxicity and futility before allocation.
4. Stage II ranks the currently admissible doses by posterior regular efficacy and returns one next dose, using the prespecified lower-dose tie-break. Any increase above the current dose is treated as escalation for the `n_eval` gate.
5. The two-stage decision returns one action and one next dose; it does not return a separate retreat destination set.

7.5 Monitoring and order of operations

1. Reconstruct interim endpoint data at `t_now`.
2. Fit/update toxicity and efficacy models.
3. Update persistent overtoxicity and futility states.
4. Apply early stopping.
5. Build the algorithm-specific candidate dose set.
6. Apply the active one-stage or two-stage algorithm and return one next dose.
7. If the next dose is higher than the current dose, apply the `n_eval` escalation gate. If blocked, drop a triggering new-patient arrival without adding it to the queue, but retain a triggering retreat opportunity in the queue; keep the cohort closed.
8. Otherwise open the cohort at the returned next dose and fill it from the queue with new-patient-first priority; all assigned patients receive the same dose.

8. Public simulator inputs and outputs

8.1 Recommended public API

```
config <- aide_phase12_config(
  allocation = "one_stage" | "two_stage",
  cohort_size, Nmax, s1_Max, N_s2,
  n_eval, m_U, cycle_max, start_dose,
  time = list(arrival_rate = ..., t0 = ..., T_assess = ...),
  toxicity = list(model = ..., target = ..., priors = ..., mcmc = ...),
  efficacy = list(model = ..., priors = ..., mcmc = ...),
  monitoring = list(...),
  utility = list(...),
  recycle = list(priority = "new_first")
)

scenario <- aide_phase12_scenario(
  p_true, e_true,
  toxicity_ipde_dgm = list(alpha_true = ...),
  efficacy_ipde_dgm = list(alpha_true = ...)
)

fit <- simulate_AIDE_phase_I_II_event(config, scenario, seed = 1)
```

8.2 Required top-level result

Element	Content
config, scenario	Normalized inputs actually used.
admin	One row per administration with the single assigned cohort dose.
event_log	All processed events, times, and state transitions.
queue_log	Queue entry, consideration, assignment, waiting time, and terminal status.
cohort_log	Opening/closing time, action, next dose, and slot assignments.
n_eval_log	Every proposed-escalation check, qualifying administration IDs, blocked status, trigger type, and type-specific trigger disposition.
decision_log	Every model/design decision attempt, action, next dose, and stop/no-dose outcome.
retreat_log	One row per retreat opportunity with structural eligibility, queue status, retention after any n_eval block, cohort, assigned dose, or removal reason unrelated to the n_eval block.
stage1, stage2	Stage-specific sample counts, transition, candidate sets, next-dose decisions, and summaries.
final	MTD, OBD, no-selection reason, elimination vectors, utilities, duration, sample counts.
events_remaining, queue_remaining	Unprocessed/queued items at termination.

8.3 Minimum decision-log columns

Column group	Meaning
decision_id, time, trigger_event_id	Traceability.
cohort_open_before	Confirms decisions occur only when closed.
current_dose, stage	State entering decision.
n_eval_required, n_eval_observed, evaluated_admin_ids	Recorded only when action is escalation: threshold, observed administration count, and qualifying administration IDs.
escalation_blocked, trigger_type, trigger_disposition, new_patient_dropped, retreat_opportunity_retained	Audit whether n_eval blocked the proposed escalation and whether the triggering new patient was dropped before queue entry or the triggering retreat opportunity was retained in the queue.
toxicity_action, toxicity_recommended_dose	Safety recommendation.
action, next_dose, candidate_doses	Single design output controlling the entire cohort; candidate set retained only for audit.
stop_trial, stop_reason	Stopping result.
toxicity_eliminated, efficacy_futile	Persistent exclusion state.

8.4 Operating-characteristic output

- MTD and OBD selection percentages, early stopping, and no-selection percentages.
- Mean administrations and unique patients by dose and overall.
- Mean IPDE administrations by assigned dose and cycle.
- Mean trial duration and queue waiting time by event type.
- Frequency of proposed escalations and n_eval escalation blocks, with separate counts for triggering new patients dropped before queue entry and triggering retreat opportunities retained in the queue.
- Frequency of retreat structural failures and the number of eligible retreat opportunities assigned or left waiting.
- Composition of each cohort by new versus retreat patients; all patients in a cohort share one next dose.
- Optional posterior bias/RMSE/coverage based on stored scalar or vector summaries only; MCMC draws are never retained in trial outputs.

9. Migration plan

Step	Deliverable
1. Freeze scientific helpers	Copy the current CRM and four efficacy JAGS adapters into tests; verify their input/output dimensions on fixed synthetic data.
2. Build state schemas	Create constructors for config, scenario, events, queue, cohort, admin, and logs.
3. Extract decisions	Move one-stage and two-stage logic out of simulate_AIDE_phase_I_II into pure functions.

Step	Deliverable
4. Generalize model adapters	Dispatch TITE likelihoods by the selected toxicity and efficacy models; remove the fixed-r restriction; discard posterior sample matrices after summarization; return no model diagnostics.
5. Build event primitives	Port schedule/pop/fill logic from AIDE_CRM_helper_TITE.R with new-first priority and queue retention.
6. Add escalation-only n_eval gate	Count evaluated administrations only for a proposed escalation; repeated administrations from one patient count separately. When blocked, drop a triggering new arrival before queue entry but retain a triggering retreat opportunity in the queue.
7. Simplify recycling	Retain structural eligibility and new-first queue priority; remove destination evaluation and assign every cohort member to one next dose.
8. Integrate stages	Use one event loop for both one-stage and two-stage algorithms.
9. Standardize final output	Preserve current MTD/OBD fields while adding event, queue, escalation-block, new-trigger-drop, retreat-trigger-retention, and retreat-assignment logs without posterior draws or model diagnostics.
10. Compatibility wrapper	Map legacy arguments to config/scenario and warn on deprecated ipde_design, flexible_ipde, and random_crm_* names.
11. Regression simulations	Compare complete-outcome behavior intended to remain unchanged; validate model-specific TITE likelihoods, common-window timing, and pending outcomes.

9.1 Functions to deprecate or change in meaning

Existing item	Target treatment
ipde_design, flexible_ipde	Deprecated for the rebuilt event strategy because the model and one-stage/two-stage algorithm return one next dose for the entire cohort.
enrollment_scheme = ipde_first	Not used in the new-first TITE path; may remain only for legacy non-TITE comparisons.
n_eval_escalate	Rename to n_eval and apply only when the returned next dose is higher than the current dose.
drop_new_patients_when_suspended	Replace with explicit type-specific behavior: when n_eval blocks escalation, drop a triggering new-patient arrival before queue entry, retain a triggering retreat opportunity in the queue, and retain all other waiting opportunities.

Existing item	Target treatment
apply_random_crm_recycle_*	Remove. There are no separate model-specific recycle-destination gates; model outputs feed the common design algorithm.
random_crm_recycle_efficacy_delta	Remove from recycle logic. Efficacy model parameters remain part of the selected analysis model, not a separate eligibility threshold.
enroll_cohort()	Replace with open_cohort(), assign_one_slot(), and fill_open_cohort(), all using one cohort next dose.
eligible_ipde_ids()	Replace with structural retreat eligibility and queue selection only.

10. Verification and unit-test plan

Test	Expected assertion
Initialization	Starting cohort opens at start dose without applying n_eval.
Open cohort	Assignable events fill slots; model decision count does not increase until cohort closes.
Escalation n_eval block	Closed cohort + proposed escalation + evaluated count < n_eval -> no cohort opens; a triggering new patient is dropped without queue entry, whereas a triggering retreat opportunity remains queued.
Stay/de-escalation bypass	Closed cohort + stay or de-escalation -> cohort opens immediately even when evaluated count < n_eval.
Administration count	Each fully DLT-evaluated administration at the current dose counts once toward n_eval; repeated cycles from the same patient count as separate administrations.
New-first priority	Simultaneous/queued new and retreat -> new assigned first regardless of event sequence.
Retreat eligibility	Requires full DLT window, no DLT, no observed response, and cycle < cycle_max.
Common assessment window	DLT and efficacy ascertainment use the same T_assess, and retreat eligibility is evaluated at that common window end.
Single next dose	The one-stage/two-stage algorithm returns one next dose; new and retreat assignments in the cohort equal that dose.
Dose-homogeneous cohort	A mixed new/retreat cohort contains no dose mismatch across administration rows.
Retreat queue scan	An ineligible retreat does not block a later structurally eligible retreat after new-patient entries are exhausted.

Test	Expected assertion
Toxicity likelihood	For each selected toxicity model, a pending non-DLT contributes $1 - w_T \theta_T$ under its model-specific TITE probability.
Efficacy likelihood	For each selected efficacy model, a pending nonresponse contributes $w_E \log(1 - \theta_E)$ using T_{assess} .
One-stage	Direct de-escalation cannot be overridden; local utility sets match m_U rule.
Two-stage transition	Transition uses cumulative Stage I sample $\geq s1_Max$ and stay recommendation.
Monitoring	Overtoxicity updates as specified. Futility uses fully ascertained efficacy outcomes only; pending efficacy administrations are excluded and persistent elimination is applied as specified.
Stopping	Dose-1 toxicity, all efficacy-futile, administration cap, and no-assignable-dose outcomes terminate with explicit reasons.
Final selection	OBD is restricted according to the prespecified tried/admissible/MTD rules and lower-dose tie-break.
Reproducibility	Same trial seed reproduces event times, outcomes, queue order, new-trigger drops, retreat-trigger retention, and the final result.
OC wrapper	Trial i uses a deterministic seed offset; aggregate metrics match direct aggregation and no result contains posterior sample matrices or model diagnostics.

11. Finalized implementation specifications

The following implementation choices are finalized and should be implemented exactly as stated. They are no longer open defaults or confirmation items.

Item	Final specification
n_eval unit	n_eval counts fully DLT-evaluated administrations at the current dose, not unique patients. An administration qualifies when its DLT is observed by t_now or T_assess is complete without DLT. Repeated administrations from the same patient count separately.
Escalation classification	Adopt the proposed default: any returned $next_dose$ greater than $current_dose$ is an escalation, including Stage II jumps, and is subject to n_eval .
Dose-1 DGM boundary	Adopt the proposed default: define $p_0 = e_0 = 0$, so an IPDE administration at dose 1 has the same true toxicity and efficacy probabilities as a regular dose-1 administration.

Item	Final specification
TITE toxicity model files	Confirmed: every selectable discount-r or additive-alpha toxicity model must have a verified model-specific TITE implementation before simulation. Configuration validation must stop with an explicit error if the required backend is unavailable.
Stage II restrictions	Confirmed: the MTD ceiling, tried-dose restriction, no-skipping rule, and any Stage II candidate restrictions are explicit configuration fields and are applied by the two-stage algorithm.
Futility information threshold	min_eff_n_for_futility counts fully ascertained efficacy administrations only. An administration whose efficacy outcome is still pending is excluded from the futility numerator, denominator, and minimum-information count.

Implementation acceptance criteria

- The simulator has one event loop shared by one-stage and two-stage designs.
- The event engine contains no hard-coded fixed-r restriction; TITE dispatches to the toxicity and efficacy models selected in config.
- The n_eval gate is logged only for proposed escalations. Stay and de-escalation decisions proceed immediately. When an escalation is blocked, a triggering new-patient arrival is dropped before queue entry, whereas a triggering retreat opportunity remains queued for future consideration.
- New-patient-first priority is deterministic and independently testable.
- A cohort has one next dose, and both new and structurally eligible retreat patients are assigned to that dose.
- The selected toxicity model and all four efficacy models support TITE likelihood operation through standardized adapters. Posterior draws and model diagnostics are not stored in model objects, logs, trial results, or operating-characteristic outputs.
- Final outputs retain the current MTD/OBD and OC metrics and add event, queue, escalation-gate, new-patient-drop, retreat-retention, and retreat-assignment audit trails.

End of implementation blueprint